In [1]:
```r
library(Seurat)
# library(SeuratDisk)

library(reticulate)
library(anndata)

library(PRROC)
library(Matrix)
library(MASS)
library(dplyr)
library(tidyr)
library(repr)
library(reshape2)

library(UpSetR)
library(grid)
library(clustree)
library(pheatmap)

library(ggplot2)
library(ggpubr)
library(RColorBrewer)
library(cowplot)
library(hexbin)
library(scico)

library(lme4)
library(broom)
library(broom.mixed)
library(ggeffects)
library(geepack)

options(repr.plot.width=10, repr.plot.height=8)

getwd()

dataset_id <- "simulated_mm_RA"
genome_id <- "mm10"
samples <- c("all", "old", "young")
TEsubset <- "TP"

for (sample in samples) {
    dir.create(paste0("figures_", dataset_id, "_", sample))
}

colorTools <- c(
    "STARsolo_TE" = "#FFBC81",
    "STARsolo_TE_EM" = "#F3E088",
    "SoloTE_unique" = "#A4DD9B",
    "SoloTE_thr2" = "#6FC69D",
    "SoloTE_thr1" = "#4BB2BB",
    "SoloTE_thr0" = "#3989BF",
    "Stellarscope" = "#4F5D93",
    "simulated" = "grey70"
)

colorSamples <- c("old"="#7A646A",
                  "young"="#80BCEC",
                  "all"="#8190a9")

thrMinCells <- 500 * 0.05
```

```
Loading required package: SeuratObject

Loading required package: sp

‘SeuratObject’ was built under R 4.3.3 but the current version is
4.4.1; it is recomended that you reinstall ‘SeuratObject’ as the ABI
for R may have changed

‘SeuratObject’ was built with package ‘Matrix’ 1.6.4 but the current
version is 1.7.5; it is recomended that you reinstall ‘SeuratObject’ as
the ABI for ‘Matrix’ may have changed


Attaching package: ‘SeuratObject’


The following objects are masked from ‘package:base’:

    intersect, t
```

```
Attaching package: ‘anndata’


The following object is masked from ‘package:SeuratObject’:

    Layers


Loading required package: rlang

Warning message:
“package ‘rlang’ was built under R version 4.4.3”
Warning message:
“package ‘MASS’ was built under R version 4.4.3”
Warning message:
“package ‘dplyr’ was built under R version 4.4.3”

Attaching package: ‘dplyr’


The following object is masked from ‘package:MASS’:

    select


The following objects are masked from ‘package:stats’:

    filter, lag


The following objects are masked from ‘package:base’:

    intersect, setdiff, setequal, union


Warning message:
“package ‘tidyr’ was built under R version 4.4.3”

Attaching package: ‘tidyr’


The following objects are masked from ‘package:Matrix’:

    expand, pack, unpack


Warning message:
“package ‘reshape2’ was built under R version 4.4.3”

Attaching package: ‘reshape2’


The following object is masked from ‘package:tidyr’:

    smiths


Loading required package: ggraph

Loading required package: ggplot2


Attaching package: ‘ggraph’


The following object is masked from ‘package:sp’:

    geometry


Warning message:
“package ‘RColorBrewer’ was built under R version 4.4.3”
Warning message:
“package ‘cowplot’ was built under R version 4.4.3”

Attaching package: ‘cowplot’


The following object is masked from ‘package:ggpubr’:
```

```
    get_legend


Registered S3 method overwritten by 'lme4':
  method           from
  na.action.merMod car

Warning message:
“package ‘broom’ was built under R version 4.4.3”

Attaching package: ‘ggeffects’


The following object is masked from ‘package:cowplot’:

    get_title
```

'/mnt/TEdata/TEbenchmarking/Rscripts'

```
Warning message in dir.create(paste0("figures_", dataset_id, "_", sample)):
“'figures_simulated_mm_RA_all' already exists”
Warning message in dir.create(paste0("figures_", dataset_id, "_", sample)):
“'figures_simulated_mm_RA_old' already exists”
Warning message in dir.create(paste0("figures_", dataset_id, "_", sample)):
“'figures_simulated_mm_RA_young' already exists”
```

In [2]:
```
theme_paper <- function(base_size = 17, base_family = "") {
  theme_minimal(base_size = base_size, base_family = base_family) +
    theme(
      plot.title = element_text(face = "bold", size = base_size + 2, hjust = 0.5),
      axis.title = element_text(size = base_size),
      axis.text  = element_text(size = base_size * 0.9),
      legend.title = element_text(size = base_size),
      legend.text  = element_text(size = base_size * 0.9),
      panel.grid.major = element_blank(),
      panel.grid.minor = element_blank(),
      plot.margin = margin(10, 10, 10, 10)
    )
}
theme_set(theme_paper())
```

In [ ]:
```
#import workspace
load("workspaces/00_evaluation_objectCreation.Rdata")
```

In [4]:
```
# load truncation annotation generated in /mnt/TEdata/snakemake_TE_full/computeAge_mm10
annotation_truncation <- read.table("annotation/annotated_tes_mm10.tsv", header = TRUE)
```

In [8]:
```
table(annotation_truncation$status)
table(annotation_truncation$annotation)

# Full_Length – the insertion covers ≥90% of the repeat consensus sequence
# 5prime_Truncated – >10% of the consensus is missing from the 5' end only
# 3prime_Truncated – >10% of the consensus is missing from the 3' end only
# Internal_Fragment – >10% missing from both ends; only the middle of the element is present
# Truncated_Other – coverage <90% but neither end gap exceeds 10%; typically a slightly degraded or ambiguously
```

```
Full_Length   Truncated
    2448693     2885046
 3prime_Truncated  5prime_Truncated       Full_Length Internal_Fragment
           701149           1282138           2448693            826993
  Truncated_Other
            74766
```

In [9]:
```
colnames(annotation_truncation) <- gsub(colnames(annotation_truncation), pattern="genoName", replacement="chr")
colnames(annotation_truncation) <- gsub(colnames(annotation_truncation), pattern="genoStart", replacement="star
colnames(annotation_truncation) <- gsub(colnames(annotation_truncation), pattern="genoEnd", replacement="end")
annotation_truncation$start <- annotation_truncation$start + 1
colnames(annotation_truncation)
```

'bin' · 'swScore' · 'milliDiv' · 'milliDel' · 'milliIns' · 'chr' · 'start' · 'end' · 'genoLeft' · 'strand' · 'repName' · 'repClass' · 'repFamily' · 'repStart' · 'repEnd' · 'repLeft' · 'id' · 'cons_len' · 'aln_len' · 'coverage' · 'status' · 'annotation'

In [10]:
```
# merge truncation info to TE annotation
conversion_table <- merge(conversion_table,
                        annotation_truncation[,c("chr","start","end","strand","coverage","status", "annotation"
                        all.x=T)
dim(conversion_table)

# a few loci differ (7k)
sum(is.na(conversion_table$status))
```

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7134

In [11]:
```
# add age bin to conversion table
breaks <- c(0, 2, 5, 10, 15, 25, 40, 60, Inf)

conversion_table$age_bin <- cut(
  conversion_table$mya,
  breaks = breaks,
  labels = c("0-2", "2-5", "5-10", "10-15", "15-25", "25-40", "40-60", "60+"),
  right = FALSE,
  include.lowest = TRUE
)

table(conversion_table$age_bin)
```

```
   0-2     2-5    5-10   10-15   15-25   25-40   40-60     60+ 
 25622  111376  312875  265885  645109 1440279  901650   17530 
```

In [12]:
```
# add finer age bins to conversion table
breaks2 <- c(0, 1, 2, 3, 4, 5, 7, 10, 15, 25, 40, 60, Inf)

conversion_table$age_bin_finer <- cut(
  conversion_table$mya,
  breaks = breaks2,
  labels = c("0-1", "1-2", "2-3", "3-4", "4-5", "5-7", "7-10", "10-15", "15-25", "25-40", "40-60", "60+"),
  right = FALSE,
  include.lowest = TRUE
)

table(conversion_table$age_bin_finer)
```

```
   0-1     1-2     2-3     3-4     4-5     5-7    7-10   10-15   15-25   25-40 
 10081   15541   24896   40914   45566  115978  196897  265885  645109 1440279 
 40-60     60+ 
901650   17530 
```

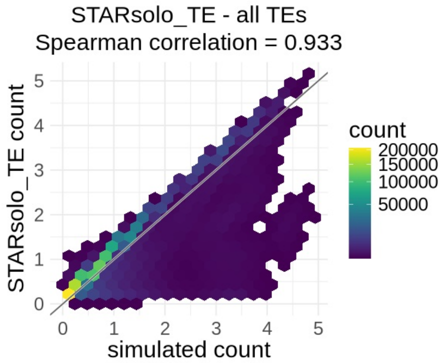
Density maps

In [12]:
```
options(repr.plot.width=5, repr.plot.height=4)

layer <- "data"

dfAgeCorr <- NULL

densityPlots <- list()
densityPlotsDf <- NULL

dfCorrList <- list()

# Get density of points in 2 dimensions.
# @param x A numeric vector.
# @param y A numeric vector.
# @param n Create a square n by n grid to compute density.
# @return The density within each square.
get_density <- function(x, y, ...) {
  dens <- MASS::kde2d(x, y,
            h = c(ifelse(bandwidth.nrd(x) == 0, 0.01, bandwidth.nrd(x)),
            ifelse(bandwidth.nrd(x) == 0, 0.01, bandwidth.nrd(y))), ...)
  ix <- findInterval(x, dens$x)
  iy <- findInterval(y, dens$y)
  ii <- cbind(ix, iy)
  return(dens$z[ii])
}
```

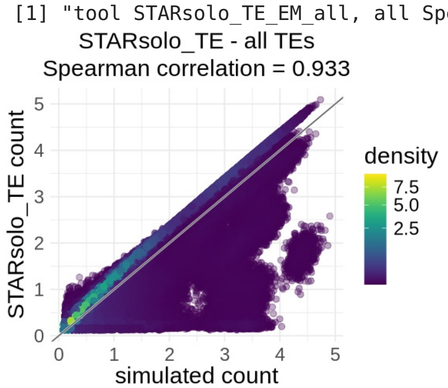
In [13]:
```
options(repr.plot.width=5, repr.plot.height=4)
#pdf("figures/densityPlots.pdf")
for(sample in samples){

    splatter_obj <- splatter_objs[[sample]]

    densityPlots[[sample]] <- list()
    dfCorrList[[sample]] <- list()

    for(tool in names(objList)){
        obj <- objList[[tool]][[sample]]

        gc()

        TP_TEs <- intersect(Features(splatter_obj),Features(obj))

        # extract matrices of TP TE loci from simulated and estimated matrices
```

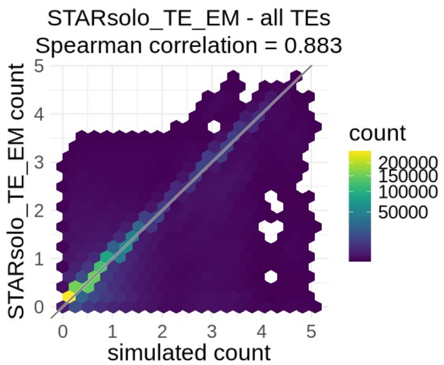
```r
        splatter_subset <- GetAssayData(splatter_obj, layer = layer)[TP_TEs,]
        tool_subset <- GetAssayData(obj, layer = layer)[TP_TEs,]

        splatter_df <- reshape2::melt(as.matrix(splatter_subset))
        tool_df <- reshape2::melt(as.matrix(tool_subset))

        df <- splatter_df
        df$tool_value <- tool_df$value
        colnames(df) <- c("TE","Cell", "sim_count", "tool_count")

        df$ageClass <- "sample"
        df$order <- conversion_table$class[match(df$TE, conversion_table$stellarscopeID)]

        df <- df[rowSums(is.na(df))==0,]

        # keep only TP (loci detected and expressed in that cell, both simulated and estimated counts > 0)
        df <- df[((df$sim_count > 0) & (df$tool_count > 0)),]

        dfCorrList[[sample]][[tool]] <- cor(df$sim_count, df$tool_count, method = "spearman")

        print(paste0("tool ", obj@project.name, ", ", sample, " Spearman correlation: ",
                    cor(df$sim_count, df$tool_count, method = "spearman")))

        densityPlots[[sample]][[tool]] <- ggplot(df, aes(x=sim_count, y=tool_count)) +
                geom_hex(bins = 20) +
                scale_fill_viridis(option = "viridis", trans = "sqrt") +
                ggtitle(label = paste0("Spearman correlation = ", round(dfCorrList[[sample]][[tool]],3))) +
                geom_abline(intercept = 0, slope = 1, color="white", linewidth=0.8) +
                geom_abline(intercept = 0, slope = 1, color="grey40", linewidth=0.5) +
                xlab("simulated count") +
                ylab(paste0(tool, " count"))+
                # ylim(c(0, max(df_young$sim_count))) +
                theme_minimal() +
                theme(text=element_text(size=18),
                plot.title = element_text(size=18, hjust=0.5),
                plot.subtitle = element_text(size=18, hjust=0.5))

        show(ggplot(df, aes(x=sim_count, y=tool_count)) +
                geom_hex(bins = 20) +
                scale_fill_viridis(option = "viridis", trans = "sqrt") +
                ggtitle(paste0(tool, " - " , sample , " TEs"),
                        subtitle = paste0("Spearman correlation = ", round(dfCorrList[[sample]][[tool]],3))) +
                geom_abline(intercept = 0, slope = 1, color="white", linewidth=0.8) +
                geom_abline(intercept = 0, slope = 1, color="grey40", linewidth=0.5) +
                xlab("simulated count") +
                ylab(paste0(tool, " count"))+
                # ylim(c(0, max (df_young$sim_count))) +
                theme_minimal() +
                theme(text=element_text(size=18),
                plot.title = element_text(size=18, hjust=0.5),
                plot.subtitle = element_text(size=18, hjust=0.5))
            )
        ggsave(paste0("figures_simulated_mm_RA_",sample,"/densityPlot_", TEsubset, "_normalized_",sample, "_",
            device="pdf", width=5, height=4)

        df$density <- get_density(df$sim_count, df$tool_count, n = 100)
        show(
            ggplot(df) +
                geom_point(aes(sim_count, tool_count, color = density, alpha = log(density)), size=2) +
                scale_color_viridis(option = "viridis", trans = "sqrt") +
                ggtitle(paste0(tool, " - " , sample , " TEs"),
                        subtitle = paste0("Spearman correlation = ", round(dfCorrList[[sample]][[tool]], 3))) +
                geom_abline(intercept = 0, slope = 1, color="white", linewidth=0.8) +
                geom_abline(intercept = 0, slope = 1, color="grey40", linewidth=0.5) +
                scale_alpha_continuous(range = c(0.3, 1)) +
                xlab("simulated count") +
                ylab(paste0(tool, " count")) +
                guides(alpha = "none") +
                theme_minimal() +
                theme(text=element_text(size=18),
                plot.title = element_text(size=18, hjust=0.5),
                plot.subtitle = element_text(size=18, hjust=0.5))
            )
        ggsave(paste0("figures_simulated_mm_RA_",sample,"/densityScatterPlot_", TEsubset, "_normalized_",sample
            device="png", width=5, height=4, dpi=600)
        ggsave(paste0("figures_simulated_mm_RA_",sample,"/densityScatterPlot_", TEsubset, "_normalized_",sample
            device="pdf", width=5, height=4)

        # create df with all the data points
        densityPlotsDf <- rbind(densityPlotsDf, cbind(df, tool, sample))
    }
}
```

```
[1] "tool STARsolo_TE_all, all Spearman correlation: 0.933443301132323"
```

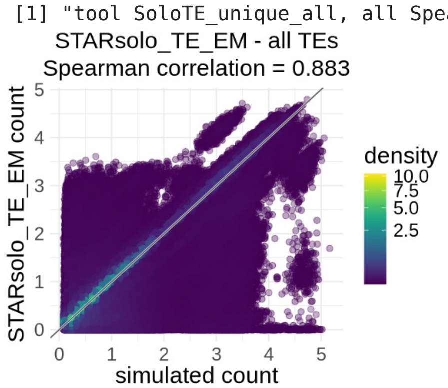


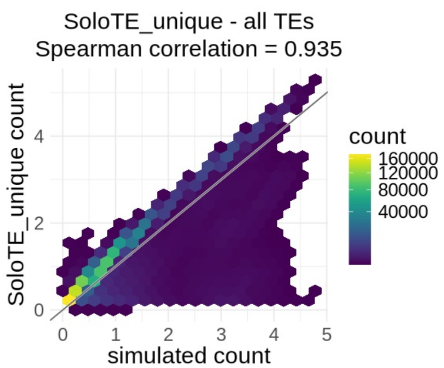
[1] "tool STARsolo_TE_EM_all, all Spearman correlation: 0.8830898019602"





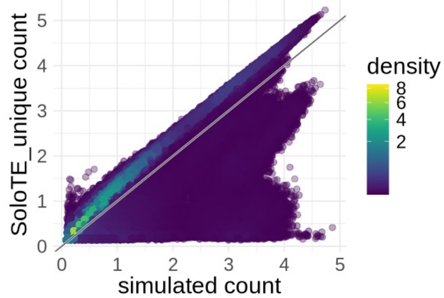
[1] "tool SoloTE_unique_all, all Spearman correlation: 0.935359928956945"



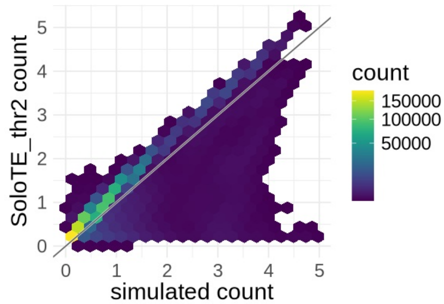


[1] "tool SoloTE_thr2_all, all Spearman correlation: 0.919210425766211"

SoloTE_unique - all TEs
Spearman correlation = 0.935

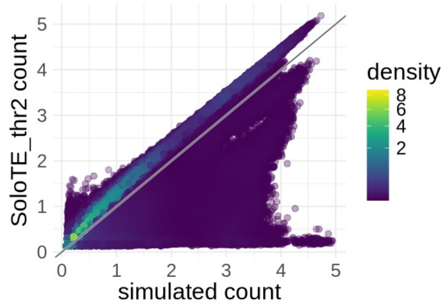


SoloTE_thr2 - all TEs
Spearman correlation = 0.919

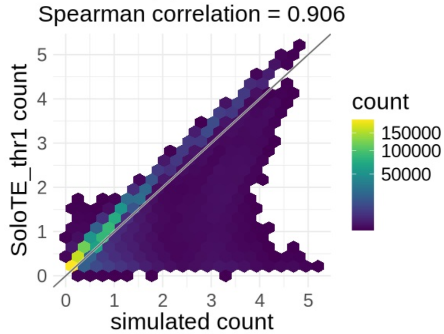



```
[1] "tool SoloTE_thr1_all, all Spearman correlation: 0.905700762486172"
```

SoloTE_thr2 - all TEs
Spearman correlation = 0.919

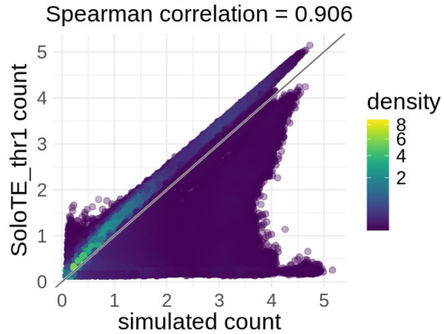


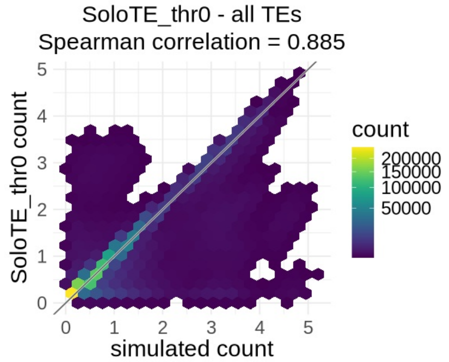
SoloTE_thr1 - all TEs
Spearman correlation = 0.906



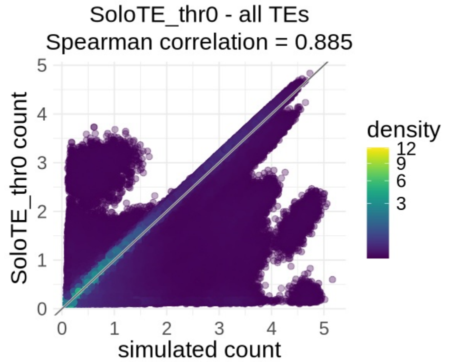

```
[1] "tool SoloTE_thr0_all, all Spearman correlation: 0.885095601154214"
```

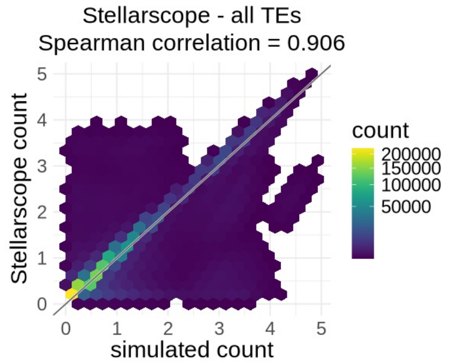
SoloTE_thr1 - all TEs
Spearman correlation = 0.906



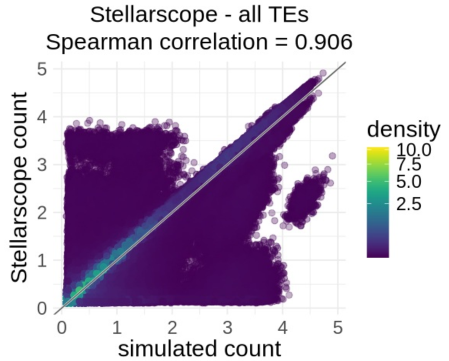


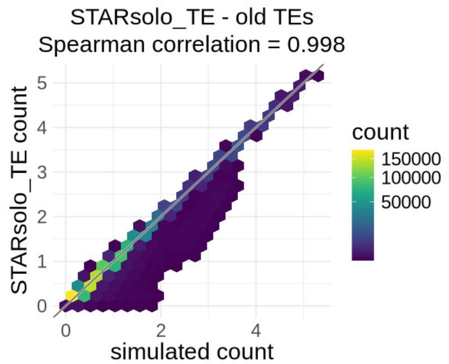
[1] "tool Stellarscope_all, all Spearman correlation: 0.905501562193628"



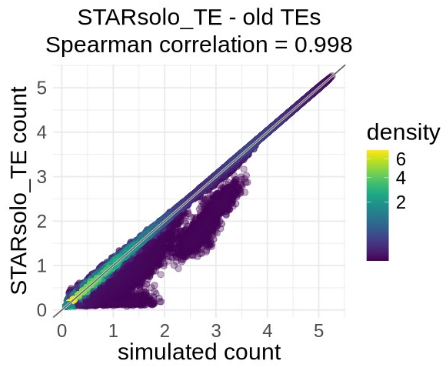


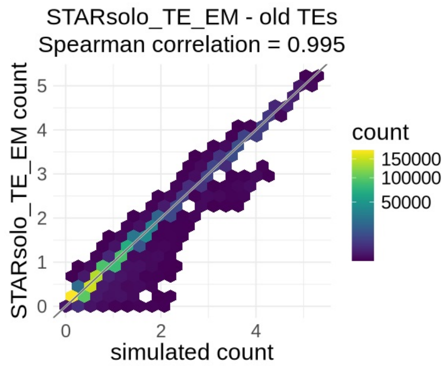
[1] "tool STARsolo_TE_old, old Spearman correlation: 0.997653120212236"



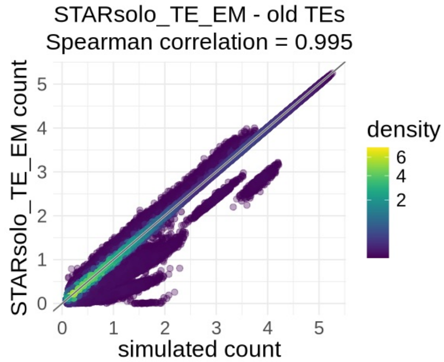


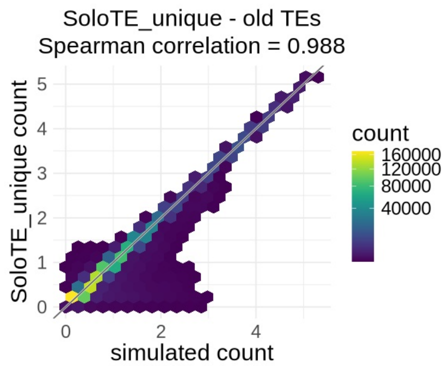
[1] "tool STARsolo_TE_EM_old, old Spearman correlation: 0.995484431448249"



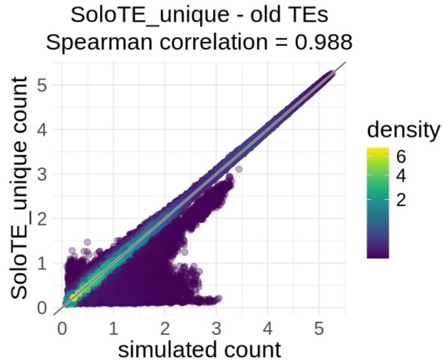



```
[1] "tool SoloTE_unique_old, old Spearman correlation: 0.988217783655751"
```

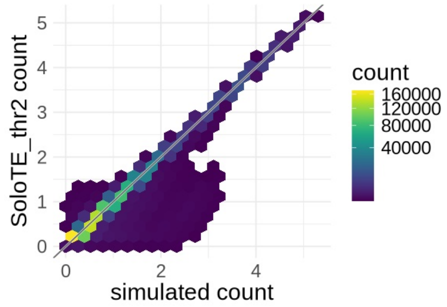





```
[1] "tool SoloTE_thr2_old, old Spearman correlation: 0.990095916148809"
```

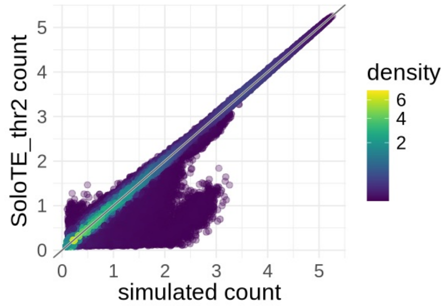


SoloTE_thr2 - old TEs
Spearman correlation = 0.99

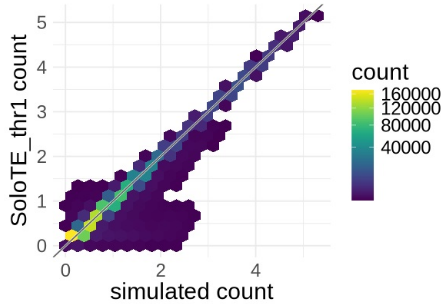


[1] "tool SoloTE_thr1_old, old Spearman correlation: 0.990762869843309"

SoloTE_thr2 - old TEs
Spearman correlation = 0.99

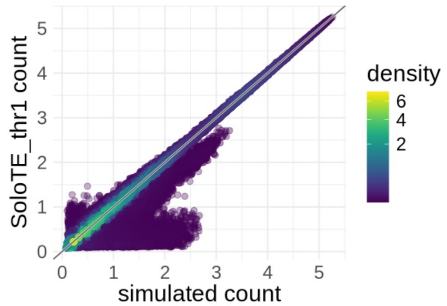


SoloTE_thr1 - old TEs
Spearman correlation = 0.991

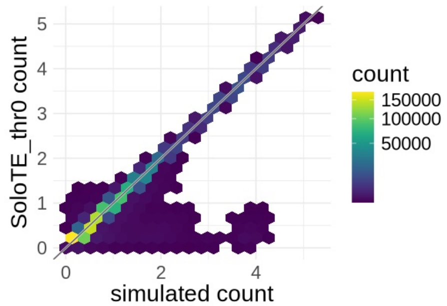


[1] "tool SoloTE_thr0_old, old Spearman correlation: 0.990067983777016"

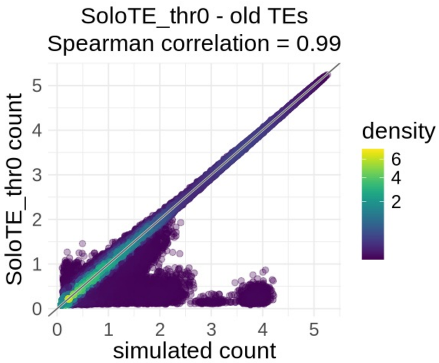
SoloTE_thr1 - old TEs
Spearman correlation = 0.991

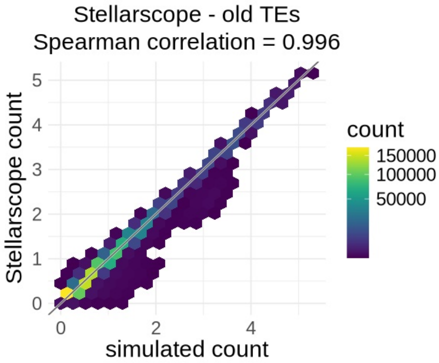


SoloTE_thr0 - old TEs
Spearman correlation = 0.99

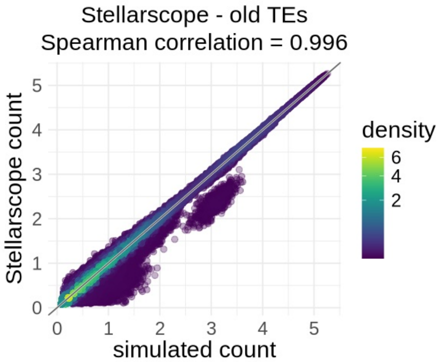


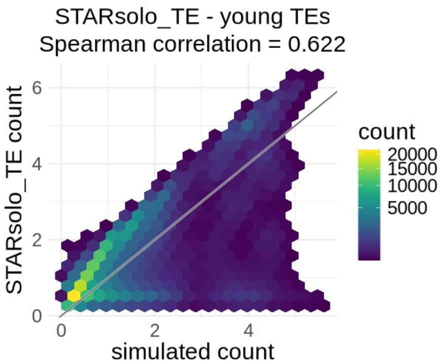
[1] "tool Stellarscope_old, old Spearman correlation: 0.995638220833332"



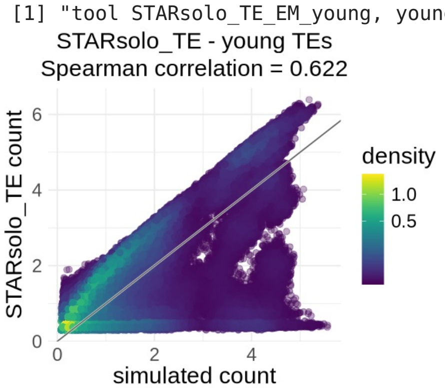


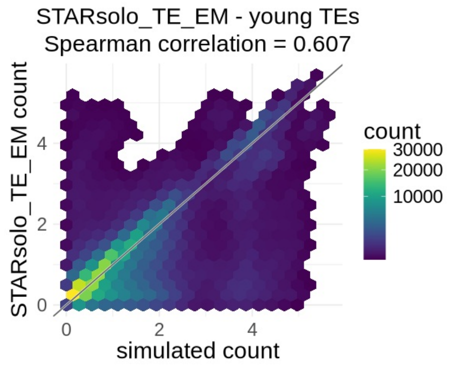

```
[1] "tool STARsolo_TE_young, young Spearman correlation: 0.62155998136769"
```



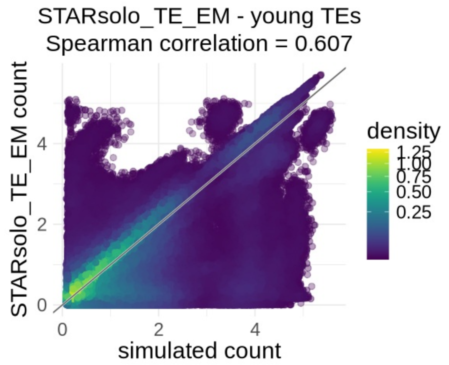


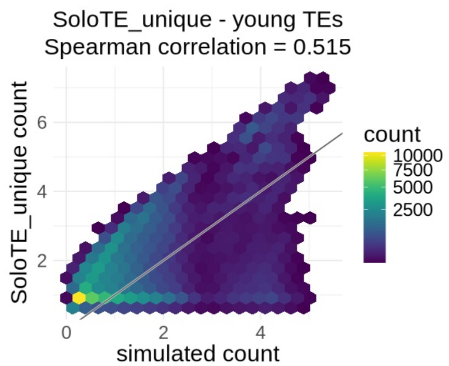

```
[1] "tool STARsolo_TE_EM_young, young Spearman correlation: 0.60735435625613"
```



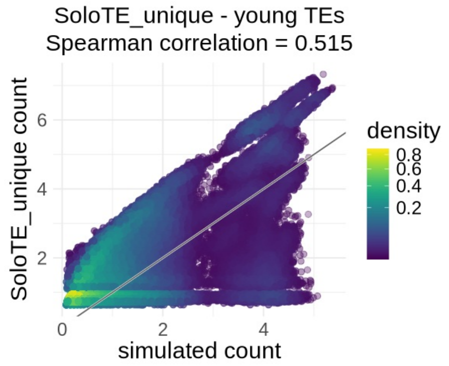


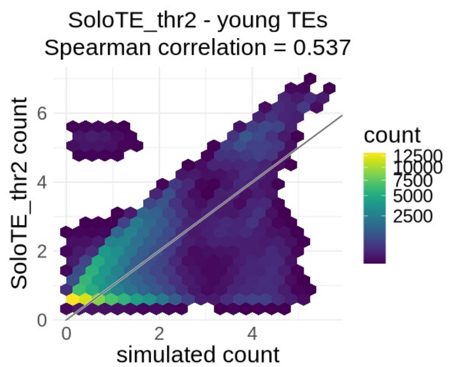
[1] "tool SoloTE_unique_young, young Spearman correlation: 0.515176528086933"



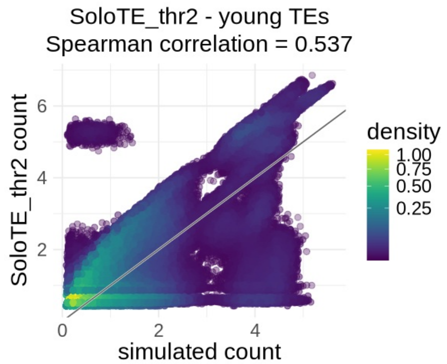


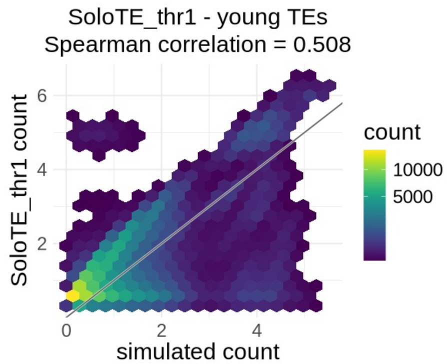
[1] "tool SoloTE_thr2_young, young Spearman correlation: 0.536590551669974"



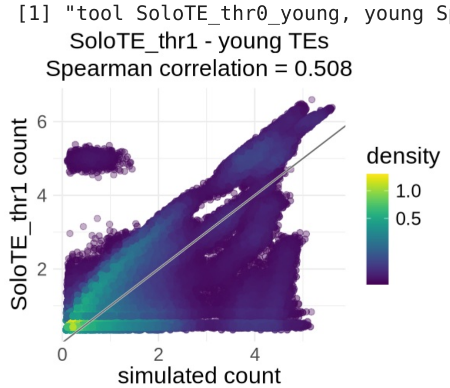


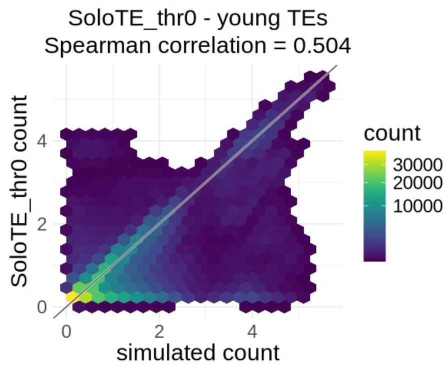
[1] "tool SoloTE_thr1_young, young Spearman correlation: 0.508296867610954"



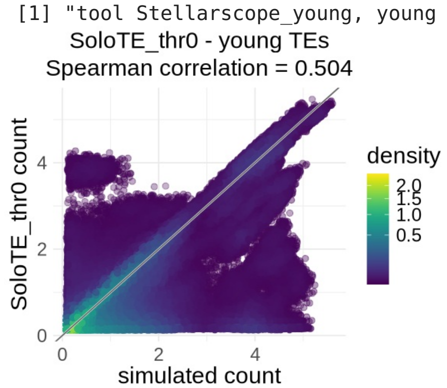


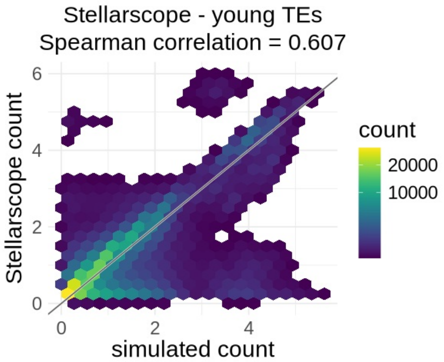

```
[1] "tool SoloTE_thr0_young, young Spearman correlation: 0.503731991978124"
```

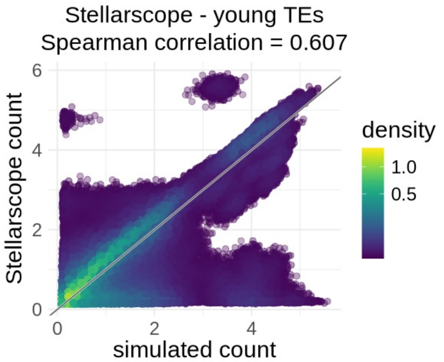





```
[1] "tool Stellarscope_young, young Spearman correlation: 0.607237446000832"
```







In [14]:
```
save.image(paste0("workspaces/", dataset_id, "_levels_density_maps_", TEsubset,".Rdata"))
```

Checkpoint

In [5]:
```
load(paste0("workspaces/", dataset_id, "_levels_density_maps_", TEsubset,".Rdata"))
```

In [13]:
```
gc()
```

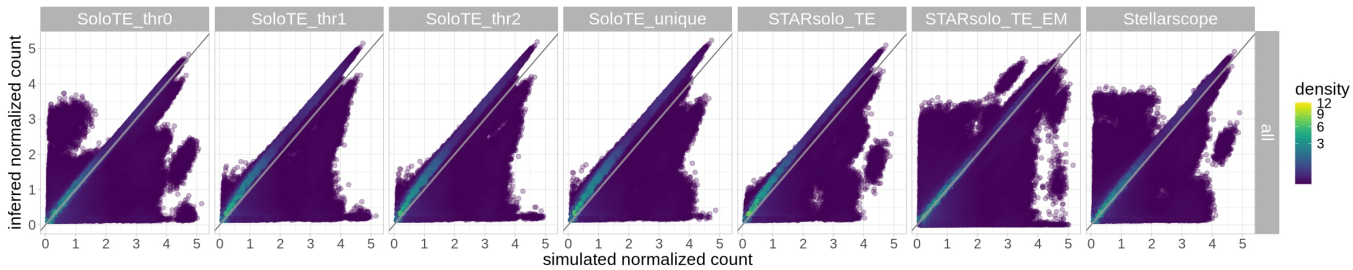
A matrix: 2 × 6 of type dbl

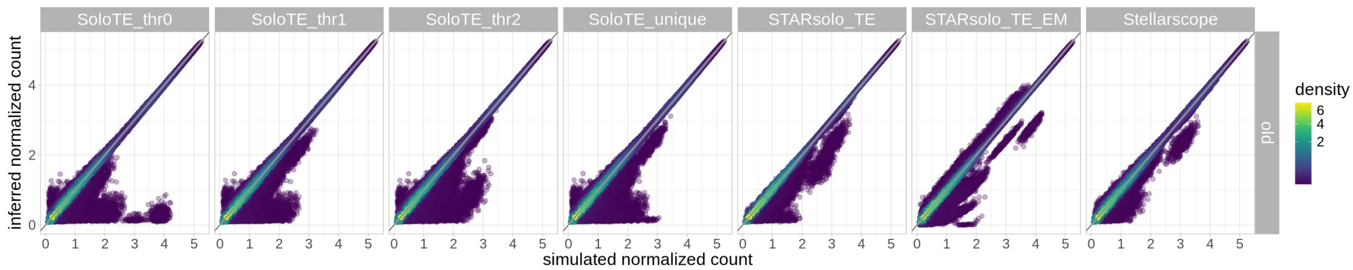
	used	(Mb)	gc trigger	(Mb)	max used	(Mb)
Ncells	19796400	1057.3	55468031	2962.4	38715006	2067.7
Vcells	607876933	4637.8	879731860	6711.9	879471327	6709.9

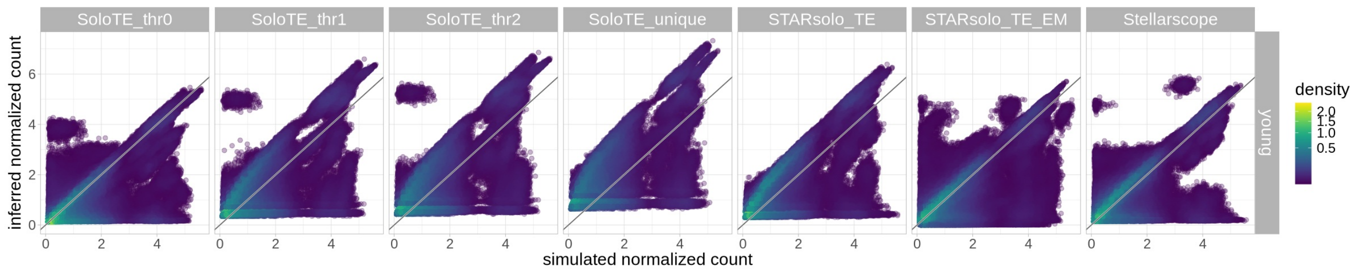
In [3]:
```
options(repr.plot.width=20, repr.plot.height=4)
gc()

for(sample in samples){
    show(
        ggplot(densityPlotsDf[densityPlotsDf$sample==sample,]) +
            geom_point(aes(sim_count, tool_count, color = density, alpha=density), size=2) +
            facet_grid(rows = vars(sample), cols = vars(tool)) +
            scale_color_viridis(option = "viridis", trans = "sqrt") +
            geom_abline(intercept = 0, slope = 1, color="white", linewidth=0.8) +
            geom_abline(intercept = 0, slope = 1, color="grey40", linewidth=0.5) +
            scale_alpha_continuous(range = c(0.3, 1)) +
            xlab("simulated normalized count") +
            ylab(paste0("inferred normalized count")) +
            guides(alpha = "none") +
            theme_light() +
            theme(text=element_text(size=18), strip.text=element_text(size=18))
    )
    ggsave(paste0("figures_simulated_mm_RA_",sample,"/grid_scatter_density_plots_facet_", TEsubset,".pdf"), dev
}
```

A matrix: 2 × 6 of type dbl

	used	(Mb)	gc trigger	(Mb)	max used	(Mb)
Ncells	41721799	2228.2	76981841	4111.3	41892245	2237.3
Vcells	1035044330	7896.8	1421202610	10843.0	1035168221	7897.8





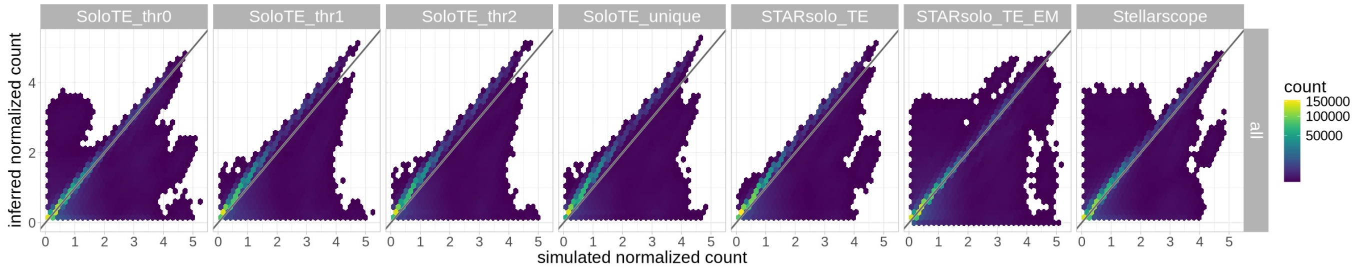


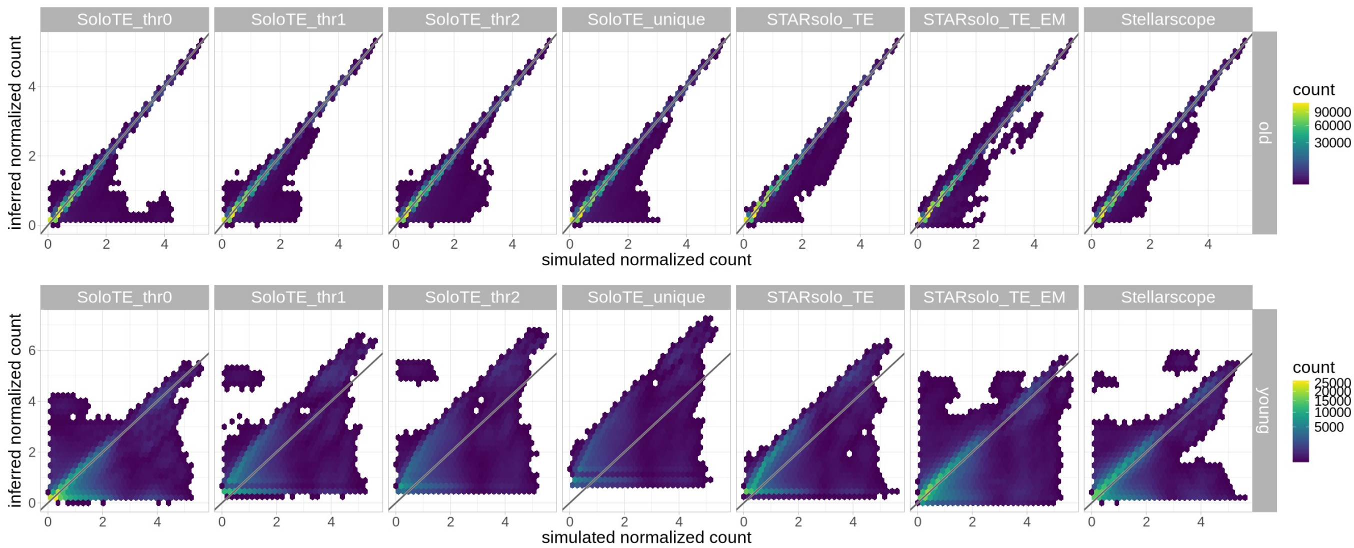
In [4]:

```
options(repr.plot.width=20, repr.plot.height=4)
gc()

for(sample in samples){
    show(
        ggplot(densityPlotsDf[densityPlotsDf$sample==sample,], aes(x=sim_count, y=tool_count)) +
            facet_grid(rows = vars(sample), cols = vars(tool)) +
            geom_hex(bins = 30) +
            scale_fill_viridis(option = "viridis", trans="sqrt") +
            #ggtitle(paste0(tool, " - " , sample , " TEs")) +
            geom_abline(intercept = 0, slope = 1, color="white", linewidth=1) +
            geom_abline(intercept = 0, slope = 1, color="grey40", linewidth=0.8) +
            xlab("simulated normalized count") +
            ylab(paste0("inferred normalized count"))+
            theme_light() +
            theme(text=element_text(size=18), strip.text=element_text(size=18))
    )
    ggsave(paste0("figures_simulated_mm_RA_",sample,"/grid_density_hex_plots_facet_",TEsubset,".pdf"), device="
}
```

A matrix: 2 × 6 of type dbl

	used	(Mb)	gc trigger	(Mb)	max used	(Mb)
Ncells	41985461	2242.3	76981841	4111.3	46382320	2477.1
Vcells	1055710228	8054.5	2463973518	18798.7	2463685513	18796.5





Metrics by locus

In [20]:
```
densityPlotsDf$family <- conversion_table$family[match(densityPlotsDf$TE, conversion_table$stellarscopeID)]
densityPlotsDf$subfamily <- conversion_table$subfamily[match(densityPlotsDf$TE, conversion_table$stellarscopeID
densityPlotsDf$propSubs <- conversion_table$propSubs[match(densityPlotsDf$TE, conversion_table$stellarscopeID)]
```

Compute correlations

In [27]:
```
library(dplyr)

# group by TE and by tool and compute spearman correlation, only for loci detected in at least 3 cells

cor_by_TE <- densityPlotsDf %>%
  group_by(TE, tool, sample) %>%
  filter(n() >= thrMinCells) %>%
  summarise(
    order  = first(order),
    family = first(family),
    subfamily = first(subfamily),
    n = n(),
    cor = cor(sim_count, tool_count, method="spearman", use = "complete.obs"),
    p_value = cor.test(sim_count, tool_count, method = "spearman", use = "complete.obs")$p.value,
    .groups = "drop"
  )
```

```
Warning message:
“There were 19698 warnings in `summarise()`.
The first warning was:
ℹ In argument: `p_value = cor.test(sim_count, tool_count, method = "spearman",
  use = "complete.obs")$p.value`.
ℹ In group 2: `TE = Lx4B-dup3757`, `tool = "STARsolo_TE"`, `sample = "old"`.
Caused by warning in `cor.test.default()`:
! Cannot compute exact p-value with ties
ℹ Run `dplyr::last_dplyr_warnings()` to see the 19697 remaining warnings.”
```

In [29]:
```
# order tools to plot
cor_by_TE$tool <- factor(cor_by_TE$tool , levels = rev(names(colorTools)))

# add other covariates
cor_by_TE$mya <- conversion_table$mya[match(cor_by_TE$TE, conversion_table$stellarscopeID)]
cor_by_TE$propSubs <- conversion_table$propSubs[match(cor_by_TE$TE, conversion_table$stellarscopeID)]
cor_by_TE$TElength <- conversion_table$end[match(cor_by_TE$TE, conversion_table$stellarscopeID)] -
                                        conversion_table$start[match(cor_by_TE$TE, conversion_table$stellarscop

cor_by_TE$age_bin <- conversion_table$age_bin[match(cor_by_TE$TE, conversion_table$stellarscopeID)]
cor_by_TE$age_bin_finer <- conversion_table$age_bin_finer[match(cor_by_TE$TE, conversion_table$stellarscopeID)]
cor_by_TE$truncationStatus <- conversion_table$status[match(cor_by_TE$TE, conversion_table$stellarscopeID)]
cor_by_TE$truncationAnnotation <- conversion_table$annotation[match(cor_by_TE$TE, conversion_table$stellarscope
```

```
head(cor_by_TE)
```

A tibble: 6 × 16

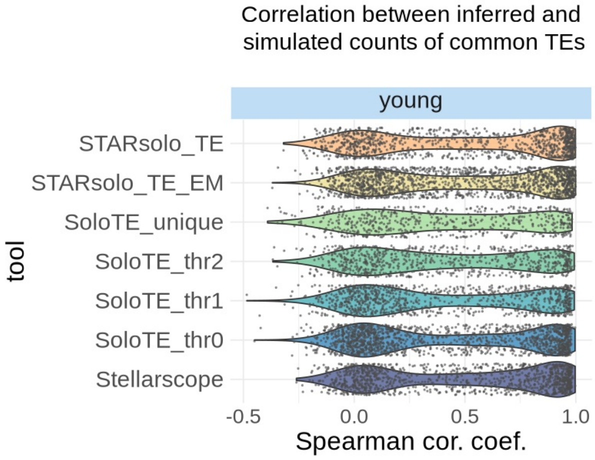
TE	tool	sample	order	family	subfamily	n	cor	p_value	mya	propSubs	TE
<fct>	<fct>	<chr>	<chr>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	
Lx4B-dup3757	STARsolo_TE	all	LINE	L1	Lx4B	357	0.9994506	0.000000e+00	25.54376	0.198	
Lx4B-dup3757	STARsolo_TE	old	LINE	L1	Lx4B	438	0.9989912	0.000000e+00	25.54376	0.198	
Lx4B-dup3757	STARsolo_TE_EM	all	LINE	L1	Lx4B	357	0.9992175	0.000000e+00	25.54376	0.198	
Lx4B-dup3757	STARsolo_TE_EM	old	LINE	L1	Lx4B	438	0.9989844	0.000000e+00	25.54376	0.198	
Lx4B-dup3757	SoloTE_thr0	all	LINE	L1	Lx4B	357	0.9794752	5.040805e-249	25.54376	0.198	
Lx4B-dup3757	SoloTE_thr0	old	LINE	L1	Lx4B	438	0.9879068	0.000000e+00	25.54376	0.198	

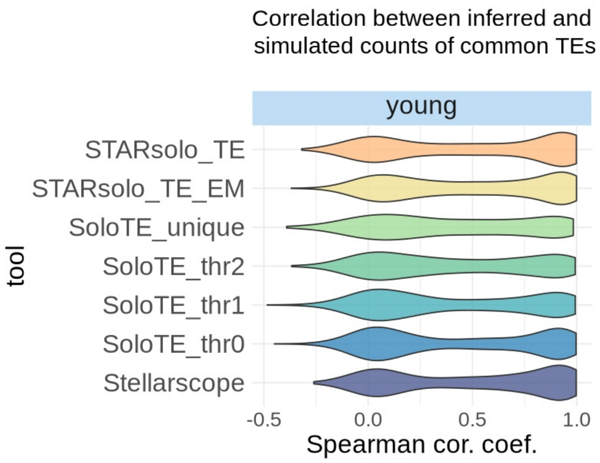
In [30]:

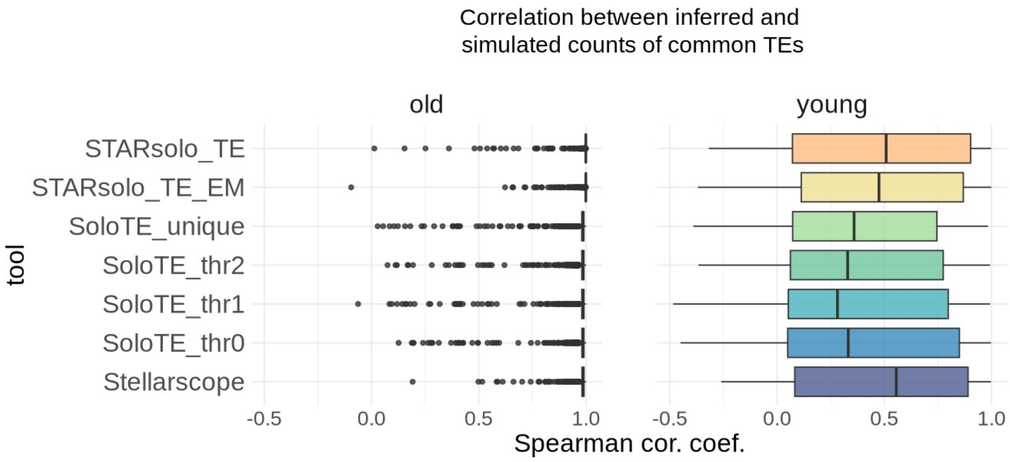
```
#library(ggbeeswarm)

options(repr.plot.width=6.5, repr.plot.height=5)

sample <- "young"

ggplot(cor_by_TE[cor_by_TE$sample == sample,]) + aes(x=cor, fill=tool,  y=tool) +
  facet_wrap(~sample) +
  geom_violin( linewidth = 0.5, alpha = 0.8 ) +
  geom_point(position = position_jitter(), size=0.3, alpha=0.5, color="grey30") +
#   geom_beeswarm(size=0.05, alpha = 0.5) +
  # geom_dotplot(binaxis= "x",
  #              stackdir = "center",
  #              dotsize = 0.05,
  #              fill = 1) +
  ggtitle("Correlation between inferred and",
          subtitle = " simulated counts of common TEs\n") +
  xlab("Spearman cor. coef.") +
  scale_fill_manual(values = colorTools) +
  theme_minimal() +
  theme(text=element_text(size=20), axis.text.y = element_text(size=18),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        strip.text = element_text(size=18, hjust=0.5, vjust = 1 ),
        strip.background = element_rect(fill=alpha(colorSamples[sample], 0.5), color = "white"),
        panel.spacing = unit(3, "lines")) + NoLegend()

ggsave(paste0("figures_simulated_mm_RA_",sample,"/correlation_perTE_violin_jitter",sample,"_byTool.pdf"))

ggplot(cor_by_TE[cor_by_TE$sample == sample,]) + aes(x=cor, fill=tool,  y=tool) +
  facet_wrap(~sample) +
  geom_violin( linewidth = 0.5, alpha = 0.8 ) +
  ggtitle("Correlation between inferred and",
          subtitle = " simulated counts of common TEs\n") +
  xlab("Spearman cor. coef.") +
  scale_fill_manual(values = colorTools) +
  theme_minimal() +
  theme(text=element_text(size=20), axis.text.y = element_text(size=20),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        strip.text = element_text(size=20, hjust=0.5, vjust = 1 ),
        strip.background = element_rect(fill=alpha(colorSamples[sample], 0.5), color = "white"),
        panel.spacing = unit(3, "lines")) + NoLegend()

ggsave(paste0("figures_",dataset_id,"_",sample,"/correlation_perTE_violin_",sample,"_byTool.pdf"),width=6.5, he


options(repr.plot.width=11, repr.plot.height=5)
ggplot(cor_by_TE[cor_by_TE$sample %in% c("old","young"),]) + aes(x=cor, fill=tool,  y=tool) +
  facet_wrap(~sample) +
  geom_boxplot( linewidth = 0.5, alpha = 0.8 ) +
  ggtitle("Correlation between inferred and",
          subtitle = " simulated counts of common TEs\n") +
  xlab("Spearman cor. coef.") +
  scale_fill_manual(values = colorTools) +
  theme_minimal() +
  theme(text=element_text(size=20), axis.text.y = element_text(size=20),
        plot.title = element_text(size=18, hjust=0.5),
```

```
        plot.subtitle = element_text(size=18, hjust=0.5),
        strip.text = element_text(size=20, hjust=0.5, vjust = 1 ),
        #strip.background = element_rect(fill=alpha(colorSamples[sample], 0.5), color = "white"),
        panel.spacing = unit(3, "lines")) + NoLegend()

ggsave(paste0("figures_",dataset_id,"_",sample,"/correlation_perTE_boxplot_",sample,"_byAge_byTool.pdf"),width=
```

```
Saving 6.67 x 6.67 in image
```







In [36]:

```
#Test bimodality
library(diptest)
dip.test(cor_by_TE[cor_by_TE$sample == "young", ]$cor)
```

```
	Hartigans' dip test for unimodality / multimodality

data:  cor_by_TE[cor_by_TE$sample == "young", ]$cor
D = 0.059632, p-value < 2.2e-16
alternative hypothesis: non-unimodal, i.e., at least bimodal
```

Test predictors of quantification accuracy

In [4]:

```
# add average expression
layer <- "data"
avgExprPerExprCellList <- list()

for(sample in samples){
    splatter_obj <- splatter_objs[[sample]]
    mat_splatter <- GetAssayData(splatter_obj, layer = layer)
    # compute average expression in the cells where each TE is expressed
```

```r
    avgExpr<- rowMeans(mat_splatter) # avg across cells
    sumExpr <- rowSums(mat_splatter)
    nCellsExpr <- rowSums(mat_splatter > 0)
    avgExprPerExprCell <- sumExpr/nCellsExpr # avg for cells where the TE is expressed
    avgExprPerExprCellList[[sample]] <- avgExprPerExprCell
}
```

In [5]:
```r
cor_by_TE$tool <- factor(cor_by_TE$tool, levels=unique(cor_by_TE$tool))
table(cor_by_TE$tool)
```

```
  STARsolo_TE STARsolo_TE_EM    SoloTE_thr0    SoloTE_thr1    SoloTE_thr2
         7910           9197           8914           7730           7319
SoloTE_unique   Stellarscope
         6887           8328
```

Aggregate "other" and "internal" truncations

In [6]:
```r
table(cor_by_TE$truncationAnnotation)

cor_by_TE$truncationType <- gsub(cor_by_TE$truncationAnnotation, pattern="Internal_Fragment", replacement="Trun
table(cor_by_TE$truncationType)
```

```
3prime_Truncated  5prime_Truncated       Full_Length Internal_Fragment
           10436             13054             17651             14515
 Truncated_Other
             629
3prime_Truncated 5prime_Truncated      Full_Length  Truncated_Other
           10436            13054            17651            15144
```

GEE

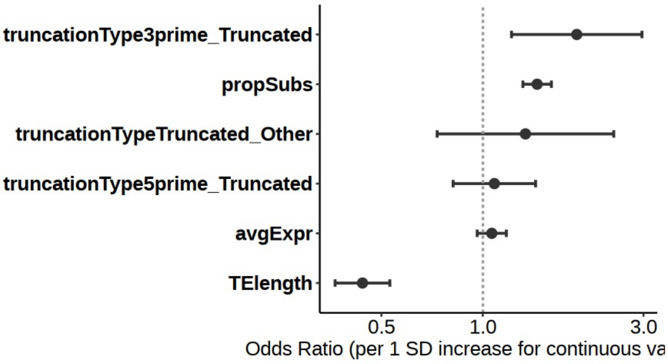
In [29]:
```r
sample <- "young"

# select correlations of young TEs
cor_by_TE_Young <- cor_by_TE[cor_by_TE$sample == sample, ]

# add avg expression
cor_by_TE_Young$avgExpr <- avgExprPerExprCellList[[sample]][as.character(cor_by_TE_Young$TE)]

# Create binary outcome
cor_by_TE_logit <- cor_by_TE_Young %>%
  mutate(high_accuracy = ifelse(cor > median(cor), 1, 0))

# scale
cor_by_TE_logit2 <- cor_by_TE_logit %>%
  mutate(
    across(c(propSubs, TElength, avgExpr), ~as.numeric(scale(.x))),
    truncationType = factor(truncationType,
      levels = c('Full_Length', '5prime_Truncated','3prime_Truncated', 'Truncated_Other'))
  )

gee_model <- geeglm(
  high_accuracy ~ tool + TElength + avgExpr + propSubs + truncationType,
  id = TE,   # cluster by locus
  data = cor_by_TE_logit2,
  family = binomial,
  corstr = "exchangeable"  # assumes equal correlation among repeated obs within a locus
)
summary(gee_model)
```

```
Call:
geeglm(formula = high_accuracy ~ tool + TElength + avgExpr +
    propSubs + truncationType, family = binomial, data = cor_by_TE_logit2,
    id = TE, corstr = "exchangeable")

 Coefficients:
                               Estimate Std.err  Wald Pr(>|W|)
(Intercept)                     -0.4311  0.0705 37.39  9.7e-10 ***
toolSTARsolo_TE_EM               0.3538  0.0448 62.28  3.0e-15 ***
toolSoloTE_thr0                  0.1266  0.0400 10.00   0.0016 **
toolSoloTE_thr1                 -0.5410  0.0605 80.08  < 2e-16 ***
toolSoloTE_thr2                 -0.7002  0.0730 92.06  < 2e-16 ***
toolSoloTE_unique               -0.9607  0.0971 97.91  < 2e-16 ***
toolStellarscope                 0.2033  0.0473 18.49  1.7e-05 ***
TElength                        -0.8226  0.0955 74.22  < 2e-16 ***
avgExpr                          0.0614  0.0508  1.46   0.2273
propSubs                         0.3715  0.0497 55.88  7.7e-14 ***
truncationType5prime_Truncated   0.0791  0.1438  0.30   0.5823
truncationType3prime_Truncated   0.6424  0.2275  7.97   0.0047 **
truncationTypeTruncated_Other    0.2916  0.3080  0.90   0.3437
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation structure = exchangeable
Estimated Scale Parameters:

            Estimate Std.err
(Intercept)     1.27   0.466
  Link = identity

Estimated Correlation Parameters:
      Estimate Std.err
alpha    0.725   0.217
Number of clusters:   1770  Maximum cluster size: 7
```

In [30]:

```r
# 1. Tidy the GEE model and manually construct CIs (broom doesn't auto-compute CIs for geeglm)
or_tab <- broom::tidy(gee_model) %>%
  mutate(
    conf.low  = estimate - 1.96 * std.error,
    conf.high = estimate + 1.96 * std.error,
    # exponentiate estimate and CI bounds to get odds ratios
    estimate  = exp(estimate),
    conf.low  = exp(conf.low),
    conf.high = exp(conf.high)
  )

# 2. Clean the dataset: remove intercept, remove tools, and clean up labels
or_plot_data <- or_tab %>%
  filter(term != "(Intercept)") %>%
  filter(!grepl("^tool", term)) %>% # Dynamically drops all tool rows
  mutate(
    # Clean up the ugly R factor prefixes for the plot labels
    term = gsub("truncationAnnotation", "", term),
    # Reorder terms based on the odds ratio estimate value
    term = reorder(term, estimate)
  )

# 3. Plotting
options(repr.plot.width=8, repr.plot.height=4)

ggplot(or_plot_data, aes(y = term, x = estimate, xmin = conf.low, xmax = conf.high)) +
  geom_point(size = 3, color = "grey20") +
  geom_errorbarh(height = 0.15, color = "grey20", linewidth = 0.8) +
  geom_vline(xintercept = 1, linetype = "dashed", color = "grey60", linewidth = 0.7) +
  scale_x_continuous(transform = "log10") +
  labs(
    x = "Odds Ratio (per 1 SD increase for continuous variables)",
    y = ""
  ) +
  theme_pubr() +
  theme(
    text = element_text(size = 15),
    axis.text.y = element_text(face = "bold")
  )

# 4. Save
ggsave(paste0("figures_simulated_mm_RA_", sample, "/OR_locus_toolfixed_GEE_features_truncationtypes.pdf"),
       device = "pdf", width = 9, height = 5)
```

```
`height` was translated to `width`.
`height` was translated to `width`.
```



Tool interaction

In [27]:

```
gee_interaction <- geeglm(
  high_accuracy ~ tool * (TElength + avgExpr + propSubs + truncationType),
  id = TE, data = cor_by_TE_logit2, family = binomial, corstr = "exchangeable"
)

QIC(gee_model)
QIC(gee_interaction)

summary(gee_interaction)
```

QIC: 10617.5102350706 **QICu:** 10573.9244318579 **Quasi Lik:** -5273.96221592894 **CIC:** 34.7929016063457 **params:** 13 **QICC:** 10617.7495513099

QIC: 10582.685831512 **QICu:** 10589.1346750002 **Quasi Lik:** -5245.56733750008 **CIC:** 45.7755782559187 **params:** 49 **QICC:** 10585.6526726987

```
Call:
geeglm(formula = high_accuracy ~ tool * (TElength + avgExpr +
    propSubs + truncationType), family = binomial, data = cor_by_TE_logit2,
    id = TE, corstr = "exchangeable")

 Coefficients:
                                                Estimate Std.err  Wald
(Intercept)                                      -0.3558  0.0673 27.94
toolSTARsolo_TE_EM                                0.3647  0.0513 50.51
toolSoloTE_thr0                                   0.1288  0.0474  7.40
toolSoloTE_thr1                                  -0.4959  0.0651 58.03
toolSoloTE_thr2                                  -0.6320  0.0802 62.06
toolSoloTE_unique                                -0.9345  0.1189 61.74
toolStellarscope                                  0.2075  0.0574 13.06
TElength                                         -0.6374  0.0859 55.12
avgExpr                                          -0.0372  0.0572  0.42
propSubs                                          0.3164  0.0569 30.90
truncationType5prime_Truncated                    0.2533  0.1565  2.62
truncationType3prime_Truncated                    0.5775  0.2441  5.60
truncationTypeTruncated_Other                    -0.1829  0.3857  0.22
toolSTARsolo_TE_EM:TElength                      -0.1075  0.0635  2.86
toolSoloTE_thr0:TElength                         -0.0125  0.0613  0.04
toolSoloTE_thr1:TElength                          0.0960  0.0680  1.99
toolSoloTE_thr2:TElength                          0.1910  0.0821  5.41
toolSoloTE_unique:TElength                        0.0847  0.1328  0.41
toolStellarscope:TElength                        -0.1057  0.0784  1.82
toolSTARsolo_TE_EM:avgExpr                        0.0693  0.0440  2.48
toolSoloTE_thr0:avgExpr                           0.0955  0.0411  5.40
toolSoloTE_thr1:avgExpr                           0.0339  0.0558  0.37
toolSoloTE_thr2:avgExpr                           0.0830  0.0622  1.78
toolSoloTE_unique:avgExpr                        -0.0609  0.0957  0.40
toolStellarscope:avgExpr                          0.1612  0.0423 14.52
toolSTARsolo_TE_EM:propSubs                       0.0514  0.0424  1.46
toolSoloTE_thr0:propSubs                          0.0406  0.0370  1.20
toolSoloTE_thr1:propSubs                         -0.0018  0.0593  0.00
toolSoloTE_thr2:propSubs                         -0.0339  0.0773  0.19
toolSoloTE_unique:propSubs                       -0.0205  0.1130  0.03
toolStellarscope:propSubs                         0.0864  0.0451  3.68
toolSTARsolo_TE_EM:truncationType5prime_Truncated -0.3150  0.1119  7.93
toolSoloTE_thr0:truncationType5prime_Truncated   -0.2081  0.0997  4.36
toolSoloTE_thr1:truncationType5prime_Truncated   -0.2899  0.2076  1.95
toolSoloTE_thr2:truncationType5prime_Truncated   -0.3846  0.2619  2.16
toolSoloTE_unique:truncationType5prime_Truncated -0.1150  0.3115  0.14
toolStellarscope:truncationType5prime_Truncated  -0.1869  0.1174  2.54
toolSTARsolo_TE_EM:truncationType3prime_Truncated -0.1106  0.1821  0.37
toolSoloTE_thr0:truncationType3prime_Truncated   -0.0236  0.1463  0.03
```

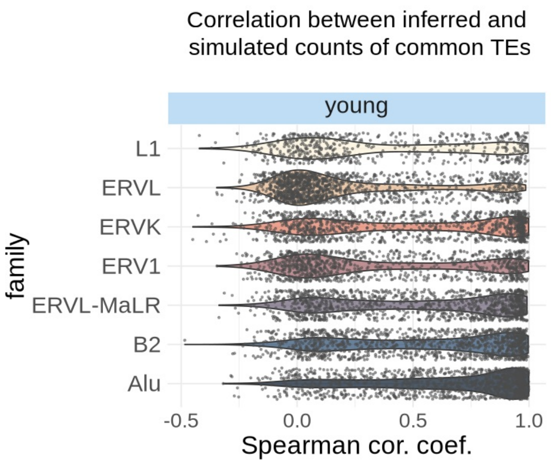
```
toolSoloTE_thr1:truncationType3prime_Truncated      0.1370  0.1897  0.52
toolSoloTE_thr2:truncationType3prime_Truncated      0.2825  0.1840  2.36
toolSoloTE_unique:truncationType3prime_Truncated    0.2327  0.2858  0.66
toolStellarscope:truncationType3prime_Truncated    -0.0764  0.1663  0.21
toolSTARsolo_TE_EM:truncationTypeTruncated_Other    0.4617  0.3862  1.43
toolSoloTE_thr0:truncationTypeTruncated_Other       0.7247  0.3667  3.91
toolSoloTE_thr1:truncationTypeTruncated_Other       0.4894  0.2221  4.86
toolSoloTE_thr2:truncationTypeTruncated_Other       0.4637  0.3348  1.92
toolSoloTE_unique:truncationTypeTruncated_Other     0.9626  0.3999  5.79
toolStellarscope:truncationTypeTruncated_Other      0.3637  0.3334  1.19
                                                   Pr(>|W|)
(Intercept)                                         1.2e-07 ***
toolSTARsolo_TE_EM                                  1.2e-12 ***
toolSoloTE_thr0                                     0.00654 **
toolSoloTE_thr1                                     2.6e-14 ***
toolSoloTE_thr2                                     3.3e-15 ***
toolSoloTE_unique                                   3.9e-15 ***
toolStellarscope                                    0.00030 ***
TElength                                            1.1e-13 ***
avgExpr                                             0.51528
propSubs                                            2.7e-08 ***
truncationType5prime_Truncated                      0.10563
truncationType3prime_Truncated                      0.01799 *
truncationTypeTruncated_Other                       0.63533
toolSTARsolo_TE_EM:TElength                         0.09064 .
toolSoloTE_thr0:TElength                            0.83860
toolSoloTE_thr1:TElength                            0.15805
toolSoloTE_thr2:TElength                            0.01999 *
toolSoloTE_unique:TElength                          0.52356
toolStellarscope:TElength                           0.17717
toolSTARsolo_TE_EM:avgExpr                          0.11494
toolSoloTE_thr0:avgExpr                             0.02014 *
toolSoloTE_thr1:avgExpr                             0.54321
toolSoloTE_thr2:avgExpr                             0.18163
toolSoloTE_unique:avgExpr                           0.52460
toolStellarscope:avgExpr                            0.00014 ***
toolSTARsolo_TE_EM:propSubs                         0.22620
toolSoloTE_thr0:propSubs                            0.27254
toolSoloTE_thr1:propSubs                            0.97575
toolSoloTE_thr2:propSubs                            0.66087
toolSoloTE_unique:propSubs                          0.85581
toolStellarscope:propSubs                           0.05507 .
toolSTARsolo_TE_EM:truncationType5prime_Truncated   0.00486 **
toolSoloTE_thr0:truncationType5prime_Truncated      0.03690 *
toolSoloTE_thr1:truncationType5prime_Truncated      0.16246
toolSoloTE_thr2:truncationType5prime_Truncated      0.14185
toolSoloTE_unique:truncationType5prime_Truncated    0.71205
toolStellarscope:truncationType5prime_Truncated     0.11134
toolSTARsolo_TE_EM:truncationType3prime_Truncated   0.54363
toolSoloTE_thr0:truncationType3prime_Truncated      0.87206
toolSoloTE_thr1:truncationType3prime_Truncated      0.47016
toolSoloTE_thr2:truncationType3prime_Truncated      0.12484
toolSoloTE_unique:truncationType3prime_Truncated    0.41546
toolStellarscope:truncationType3prime_Truncated     0.64579
toolSTARsolo_TE_EM:truncationTypeTruncated_Other    0.23185
toolSoloTE_thr0:truncationTypeTruncated_Other       0.04811 *
toolSoloTE_thr1:truncationTypeTruncated_Other       0.02753 *
toolSoloTE_thr2:truncationTypeTruncated_Other       0.16608
toolSoloTE_unique:truncationTypeTruncated_Other     0.01608 *
toolStellarscope:truncationTypeTruncated_Other      0.27538
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation structure = exchangeable
Estimated Scale Parameters:

            Estimate Std.err
(Intercept)     1.12   0.111
  Link = identity

Estimated Correlation Parameters:
      Estimate Std.err
alpha    0.712   0.064
Number of clusters:   1770  Maximum cluster size: 7
```

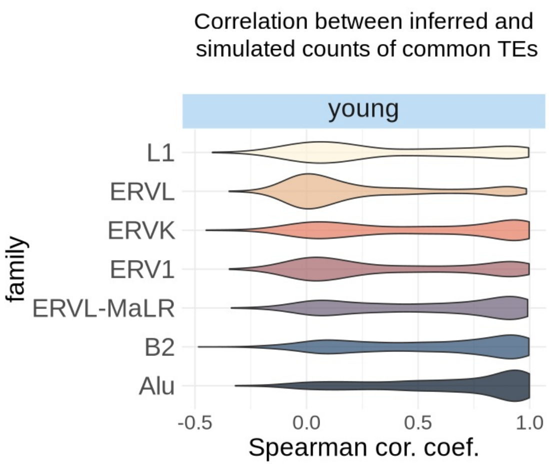
Families

In [22]:
```
unique(cor_by_TE[cor_by_TE$sample == sample,]$family)
```

'L1' · 'ERVK' · 'ERVL' · 'ERVL-MaLR' · 'B2' · 'Alu' · 'ERV1'

In [23]:
```
options(repr.plot.width=6, repr.plot.height=5)

sample <- "young"

# define order for plotting
familyOrder <- c("L1",'ERVL','ERVK','ERV1','ERVL-MaLR','B2','Alu')
df <- cor_by_TE[cor_by_TE$sample == sample,]
df$family <- factor(df$family, levels = rev(familyOrder))

ggplot(df) +
        aes(x=cor, fill=family,  y=family) +
  facet_wrap(~sample) +
  geom_violin( linewidth = 0.5, alpha = 0.7 ) +
  geom_point(position = position_jitter(), size=0.5, alpha=0.5, color="grey30") +
#   geom_beeswarm(size=0.05, alpha = 0.5) +
  # geom_dotplot(binaxis= "x",
  #              stackdir = "center",
  #              dotsize = 0.05,
  #              fill = 1) +
  ggtitle("Correlation between inferred and",
          subtitle = " simulated counts of common TEs\n") +
  xlab("Spearman cor. coef.") +
  scale_fill_scico_d(palette="lipari") +
  #scale_fill_brewer(palette="Spectral") +
  theme_minimal() +
  theme(text=element_text(size=20), axis.text.y = element_text(size=18),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        strip.text = element_text(size=18, hjust=0.5, vjust = 1 ),
        strip.background = element_rect(fill=alpha(colorSamples[sample], 0.5), color = "white"),
        panel.spacing = unit(3, "lines")) + NoLegend()

ggsave(paste0("figures_simulated_mm_RA_",sample,"/correlation_perTE_violin_jitter",sample,"_", TEsubset,".pdf")


ggplot(df) + aes(x=cor, fill=family,  y=family) +
  facet_wrap(~sample) +
  geom_violin( linewidth = 0.5, alpha = 0.7 ) +
  ggtitle("Correlation between inferred and",
          subtitle = " simulated counts of common TEs\n") +
  xlab("Spearman cor. coef.") +
  scale_fill_scico_d(palette="lipari") +
  #scale_fill_brewer(palette="Spectral") +
  theme_minimal() +
  theme(text=element_text(size=20), axis.text.y = element_text(size=20),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        strip.text = element_text(size=20, hjust=0.5, vjust = 1 ),
        strip.background = element_rect(fill=alpha(colorSamples[sample], 0.5), color = "white"),
        panel.spacing = unit(3, "lines")) + NoLegend()

ggsave(paste0("figures_",dataset_id,"_",sample,"/correlation_perTE_violin_",sample,"_", TEsubset,"_byFamily.pdf
```





Overall metrics

TP counts

In [2]:
```
# r squared
rsq <- function(x, y) summary(lm(y~x))$r.squared
rsqList <- list()


# root mean square error
rmse <- function(actual, predicted) sqrt(mean((actual - predicted)^2))
rmseList <- list()

# correlation
corList <- list()
```

In [5]:
```
layer <- "data"

TPcountsDfs <- list()

for(sample in samples){


    print(sample)

    rsqList[[sample]] <- list()
    rmseList[[sample]] <- list()
    corList[[sample]] <- list()

    splatter_obj <- splatter_objs[[sample]]

    for(tool in names(objList)){
        print(tool)

        obj <- objList[[tool]][[sample]]

        gc()

        TP_TEs <- intersect(Features(splatter_obj), Features(obj)) # TP TEs

        # extract matrices of TP TE loci from simulated and estimated matrices
        splatter_subset <- GetAssayData(splatter_obj, layer = layer)[TP_TEs,]
        tool_subset <- GetAssayData(obj, layer = layer)[TP_TEs,]
        cat("Same row names: ", identical(rownames(splatter_subset), rownames(tool_subset)), "\n")
        cat("Same col names: ", identical(colnames(splatter_subset), colnames(tool_subset)), "\n")

        print(dim(splatter_subset))
```

```r
        splatter_df <- reshape2::melt(as.matrix(splatter_subset))
        tool_df <- reshape2::melt(as.matrix(tool_subset))

        df <- splatter_df
        df$tool_value <- tool_df$value
        colnames(df) <- c("TE","Cell", "sim_count", "tool_count")

        df$ageClass <- sample

        #df <- df[rowSums(is.na(df))==0,]

        # keep only TP (loci detected and expressed in that cell, both simulated and estimated counts > 0)
        df <- df[((df$sim_count > 0) & (df$tool_count > 0)),]

        # add age and family info
        #df$ageClass <- conversion_table$ageClass[match(df$TE, conversion_table$stellarscopeID)]
        df$order <- conversion_table$class[match(df$TE, conversion_table$stellarscopeID)]
        df$family <- conversion_table$family[match(df$TE, conversion_table$stellarscopeID)]
        df$subfamily <- conversion_table$subfamily[match(df$TE, conversion_table$stellarscopeID)]

        # Compute R2
        rsqList[[sample]][[tool]] <- rsq(df$tool_count, df$sim_count)

        # Compute RMSE
        rmseList[[sample]][[tool]] <- rmse(df$sim_count, df$tool_count)

        # Compute spearman correlation
        corList[[sample]][[tool]] <- cor(df$sim_count, df$tool_count, method="spearman")

        TPcountsDfs[[sample]][[tool]] <- df
    }
}
```

```
[1] "all"
[1] "STARsolo_TE"
Same row names:  TRUE
Same col names:  TRUE
[1] 3900  500
[1] "STARsolo_TE_EM"
Same row names:  TRUE
Same col names:  TRUE
[1] 4604  500
[1] "SoloTE_unique"
Same row names:  TRUE
Same col names:  TRUE
[1] 3373  500
[1] "SoloTE_thr2"
Same row names:  TRUE
Same col names:  TRUE
[1] 3602  500
[1] "SoloTE_thr1"
Same row names:  TRUE
Same col names:  TRUE
[1] 3807  500
[1] "SoloTE_thr0"
Same row names:  TRUE
Same col names:  TRUE
[1] 4417  500
[1] "Stellarscope"
Same row names:  TRUE
Same col names:  TRUE
[1] 4114  500
[1] "old"
[1] "STARsolo_TE"
Same row names:  TRUE
Same col names:  TRUE
[1] 3001  500
[1] "STARsolo_TE_EM"
Same row names:  TRUE
Same col names:  TRUE
[1] 3028  500
[1] "SoloTE_unique"
Same row names:  TRUE
Same col names:  TRUE
[1] 2957  500
[1] "SoloTE_thr2"
Same row names:  TRUE
Same col names:  TRUE
[1] 2970  500
[1] "SoloTE_thr1"
Same row names:  TRUE
Same col names:  TRUE
[1] 2980  500
[1] "SoloTE_thr0"
```

```
Same row names:  TRUE
Same col names:  TRUE
[1] 3002  500
[1] "Stellarscope"
Same row names:  TRUE
Same col names:  TRUE
[1] 3002  500
[1] "young"
[1] "STARsolo_TE"
Same row names:  TRUE
Same col names:  TRUE
[1] 1145  500
[1] "STARsolo_TE_EM"
Same row names:  TRUE
Same col names:  TRUE
[1] 1749  500
[1] "SoloTE_unique"
Same row names:  TRUE
Same col names:  TRUE
[1] 603 500
[1] "SoloTE_thr2"
Same row names:  TRUE
Same col names:  TRUE
[1] 834 500
[1] "SoloTE_thr1"
Same row names:  TRUE
Same col names:  TRUE
[1] 1054  500
[1] "SoloTE_thr0"
Same row names:  TRUE
Same col names:  TRUE
[1] 1669  500
[1] "Stellarscope"
Same row names:  TRUE
Same col names:  TRUE
[1] 1312  500
```

In [6]:

```
cor_by_tool <- densityPlotsDf %>%
  group_by(tool, sample) %>%
  filter(n() >= 3) %>%
  summarise(
    n = n(),
    cor = cor(sim_count, tool_count, method="spearman", use = "complete.obs"),
    p_value = cor.test(sim_count, tool_count, method = "spearman", use = "complete.obs")$p.value,
    .groups = "drop"
  )

cor_by_tool[cor_by_tool$sample == "young",]
```

```
Warning message:
“There were 21 warnings in `summarise()`.
The first warning was:
ℹ In argument: `p_value = cor.test(sim_count, tool_count, method = "spearman",
  use = "complete.obs")$p.value`.
ℹ In group 1: `tool = "STARsolo_TE"`, `sample = "all"`.
Caused by warning in `cor.test.default()`:
! Cannot compute exact p-value with ties
ℹ Run `dplyr::last_dplyr_warnings()` to see the 20 remaining warnings.”
```

A tibble: 7 × 5

tool	sample	n	cor	p_value
<chr>	<chr>	<int>	<dbl>	<dbl>
STARsolo_TE	young	300004	0.6215600	0
STARsolo_TE_EM	young	482352	0.6073544	0
SoloTE_thr0	young	456565	0.5037320	0
SoloTE_thr1	young	254813	0.5082969	0
SoloTE_thr2	young	199551	0.5365906	0
SoloTE_unique	young	140777	0.5151765	0
Stellarscope	young	408953	0.6072374	0

In [7]:

```
cor_by_family <- densityPlotsDf %>%
  group_by(tool, family, sample) %>%
  filter(n() >= 3) %>%
  summarise(
    n = n(),
    cor = cor(sim_count, tool_count, method="spearman", use = "complete.obs"),
    p_value = cor.test(sim_count, tool_count, method = "spearman", use = "complete.obs")$p.value,
```

```
    .groups = "drop"
  )

cor_by_family[cor_by_family$sample == "young",]
```

```
Warning message:
“There were 497 warnings in `summarise()`.
The first warning was:
ℹ In argument: `p_value = cor.test(sim_count, tool_count, method = "spearman",
  use = "complete.obs")$p.value`.
ℹ In group 1: `tool = "STARsolo_TE"`, `family = "Alu"`, `sample = "all"`.
Caused by warning in `cor.test.default()`:
! Cannot compute exact p-value with ties
ℹ Run `dplyr::last_dplyr_warnings()` to see the 496 remaining warnings.”
```

A tibble: 49 × 6

tool	family	sample	n	cor	p_value
<chr>	<chr>	<chr>	<int>	<dbl>	<dbl>
STARsolo_TE	Alu	young	59051	0.7827778	0.000000e+00
STARsolo_TE	B2	young	56664	0.6754111	0.000000e+00
STARsolo_TE	ERV1	young	33996	0.5801418	0.000000e+00
STARsolo_TE	ERVK	young	37826	0.6815671	0.000000e+00
STARsolo_TE	ERVL	young	40723	0.1599215	2.022051e-231
STARsolo_TE	ERVL-MaLR	young	53811	0.7716902	0.000000e+00
STARsolo_TE	L1	young	17933	0.4398626	0.000000e+00
STARsolo_TE_EM	Alu	young	87732	0.7895899	0.000000e+00
STARsolo_TE_EM	B2	young	90736	0.7265592	0.000000e+00
STARsolo_TE_EM	ERV1	young	60331	0.4571532	0.000000e+00
STARsolo_TE_EM	ERVK	young	71550	0.6543664	0.000000e+00
STARsolo_TE_EM	ERVL	young	62179	0.2940367	0.000000e+00
STARsolo_TE_EM	ERVL-MaLR	young	78527	0.6894352	0.000000e+00
STARsolo_TE_EM	L1	young	31297	0.3681845	0.000000e+00
SoloTE_thr0	Alu	young	75159	0.6848052	0.000000e+00
SoloTE_thr0	B2	young	82582	0.5456561	0.000000e+00
SoloTE_thr0	ERV1	young	68538	0.4036588	0.000000e+00
SoloTE_thr0	ERVK	young	62086	0.5739313	0.000000e+00
SoloTE_thr0	ERVL	young	67106	0.1626789	0.000000e+00
SoloTE_thr0	ERVL-MaLR	young	74235	0.6686109	0.000000e+00
SoloTE_thr0	L1	young	26859	0.3285386	0.000000e+00
SoloTE_thr1	Alu	young	41874	0.6573488	0.000000e+00
SoloTE_thr1	B2	young	42825	0.6015821	0.000000e+00
SoloTE_thr1	ERV1	young	30694	0.5646061	0.000000e+00
SoloTE_thr1	ERVK	young	39142	0.5528350	0.000000e+00
SoloTE_thr1	ERVL	young	45875	0.1102929	4.089118e-124
SoloTE_thr1	ERVL-MaLR	young	43580	0.6279246	0.000000e+00
SoloTE_thr1	L1	young	10823	0.4073858	0.000000e+00
SoloTE_thr2	Alu	young	35020	0.7320329	0.000000e+00
SoloTE_thr2	B2	young	34351	0.5989238	0.000000e+00
SoloTE_thr2	ERV1	young	23181	0.6064060	0.000000e+00
SoloTE_thr2	ERVK	young	31859	0.5768421	0.000000e+00
SoloTE_thr2	ERVL	young	32885	0.1347456	4.753988e-133
SoloTE_thr2	ERVL-MaLR	young	34289	0.5943014	0.000000e+00
SoloTE_thr2	L1	young	7966	0.4082691	1.012143e-317
SoloTE_unique	Alu	young	30115	0.6568858	0.000000e+00
SoloTE_unique	B2	young	26040	0.5736877	0.000000e+00

SoloTE_unique	ERV1	young	14098	0.6537829	0.000000e+00
SoloTE_unique	ERVK	young	17928	0.4661957	0.000000e+00
SoloTE_unique	ERVL	young	22165	0.1926227	3.028889e-184
SoloTE_unique	ERVL-MaLR	young	24908	0.5681874	0.000000e+00
SoloTE_unique	L1	young	5523	0.3277011	2.085618e-138
Stellarscope	Alu	young	83685	0.7763868	0.000000e+00
Stellarscope	B2	young	81167	0.6846053	0.000000e+00
Stellarscope	ERV1	young	39424	0.5410292	0.000000e+00
Stellarscope	ERVK	young	52248	0.6884483	0.000000e+00
Stellarscope	ERVL	young	54872	0.2269728	0.000000e+00
Stellarscope	ERVL-MaLR	young	72620	0.6831396	0.000000e+00
Stellarscope	L1	young	24937	0.3393984	0.000000e+00

In [9]:

```
summary(cor_by_tool[cor_by_tool$sample == "young","cor"])
summary(cor_by_TE[cor_by_TE$sample == "young","cor"])
```

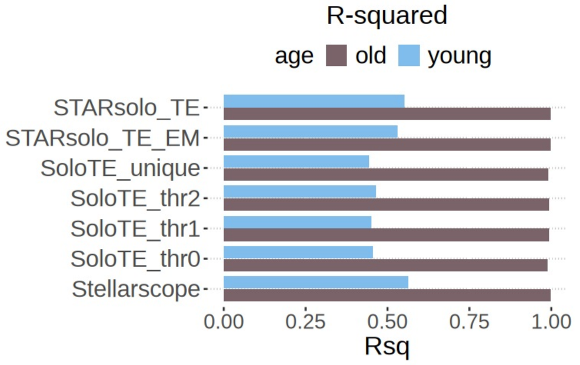
```
     cor
Min.   :0.5037
1st Qu.:0.5117
Median :0.5366
Mean   :0.5571
3rd Qu.:0.6073
Max.   :0.6216
     cor
Min.   :-0.48440
1st Qu.: 0.07393
Median : 0.42096
Mean   : 0.44382
3rd Qu.: 0.85320
Max.   : 0.99897
```

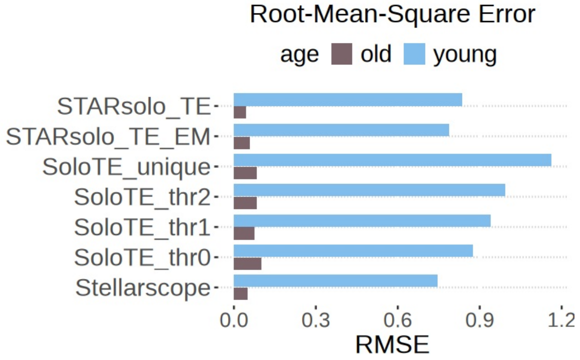
In [14]:

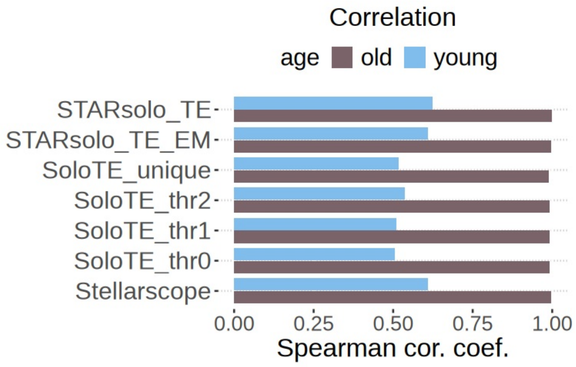
```
options(repr.plot.width=6.5, repr.plot.height=4)
colorSamples <- c("old"="#7A646A",
                "young"="#80BCEC",
                "all"="#8190a9")

# plot R squared
df_old <- cbind(melt(rsqList[["old"]]),"old")
colnames(df_old) <- c("Rsq","tool","age")
df_young <- cbind(cbind(melt(rsqList[["young"]]), "young"))
colnames(df_young) <- c("Rsq","tool","age")
df <- rbind(df_old, df_young)
df$tool <- factor(df$tool, levels=rev(names(colorTools)))

ggplot(df, aes(x=Rsq, y=tool, fill=age)) +
   geom_col(
    position = position_dodge2(width = 0.9, padding = 0.1),
    width = 0.85,
    alpha = 1
  ) +
  ylab("") +
  scale_fill_manual(values=colorSamples) +
  theme_pubclean() + ggtitle(paste0("R-squared")) +
  theme(text=element_text(size=20),
        axis.text.y = element_text(size=18),
        axis.label = element_text(size=18),
        legend.title = element_text(size=18),
        legend.text = element_text(size=18),
        plot.title = element_text(size=20, hjust=0.5))
ggsave(paste0("figures_", dataset_id, "/rsquared_barplot_",TEsubset,".pdf"), device="pdf", width=6.5, height=4)

# plot root Mean Square Error
df_old <- cbind(melt(rmseList[["old"]]),"old")
colnames(df_old) <- c("RMSE","tool","age")
df_young <- cbind(cbind(melt(rmseList[["young"]]), "young"))
colnames(df_young) <- c("RMSE","tool","age")
df <- rbind(df_old, df_young)
df$tool <- factor(df$tool, levels=rev(names(colorTools)))

ggplot(df, aes(x=RMSE, y=tool, fill=age)) +
   geom_col(
    position = position_dodge2(width = 0.9, padding = 0.1),
    width = 0.85,
    alpha = 1
```

```r
  ) +
  ylab("") +
  scale_fill_manual(values=colorSamples) +
  theme_pubclean() +
  ggtitle(paste0("Root-Mean-Square Error")) +
    theme(text=element_text(size=20),
        axis.text.y = element_text(size=19),
        legend.title = element_text(size=18),
        legend.text = element_text(size=18),
        plot.title = element_text(size=20, hjust=0.5))
ggsave(paste0("figures_", dataset_id, "/rmse_barplot_",TEsubset,".pdf"), device="pdf", width=6.5, height=4)


# plot spearman correlation
df_old <- cbind(melt(corList[["old"]]),"old")
colnames(df_old) <- c("Correlation","tool","age")
df_young <- cbind(cbind(melt(corList[["young"]]), "young"))
colnames(df_young) <- c("Correlation","tool","age")
df <- rbind(df_old, df_young)
df$tool <- factor(df$tool, levels=rev(names(colorTools)))

ggplot(df, aes(x=Correlation, y=tool, fill=age)) +
   geom_col(
    position = position_dodge2(width = 0.9, padding = 0.1),
    width = 0.85,
    alpha = 1
  ) +
  ylab("") +
  scale_fill_manual(values=colorSamples) +
  theme_pubclean() +
  ggtitle(paste0("Correlation")) +
  xlab("Spearman cor. coef.") +
    theme(text=element_text(size=20),
        axis.text.y = element_text(size=19),
        legend.title = element_text(size=18),
        legend.text = element_text(size=18),
        plot.title = element_text(size=20, hjust=0.5))

ggsave(paste0("figures_", dataset_id, "/spearmanCor_barplot_",TEsubset,".pdf"), device="pdf", width=6.5, height

write.table(df, paste0("data/", dataset_id, "_correlation_TPcounts.tsv"))
```

```
Warning message in plot_theme(plot):
“The `axis.label` theme element is not defined in the element hierarchy.”
Warning message in plot_theme(plot):
“The `axis.label` theme element is not defined in the element hierarchy.”
Warning message in plot_theme(plot):
“The `axis.label` theme element is not defined in the element hierarchy.”
```






Correlation
age
old
young
STARsolo_TE
STARsolo_TE_EM
SoloTE_unique
SoloTE_thr2
SoloTE_thr1
SoloTE_thr0
Stellarscope
0.00
0.25
0.50
0.75
1.00
Spearman cor. coef.